Beyond Constant Volatility: Integrating Jump-Diffusion and Advanced Integration Techniques into the Black-Scholes Framework for an Option Pricing model

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1. Introduction

The classic Black-Scholes-Merton model fundamentally transformed financial markets by introducing a rigorous, closed-form solution for European option pricing. However, its core assumption of constant volatility often fails to align with the empirical reality of financial markets, where phenomena such as volatility smiles and sudden asset price jumps are prevalent. As algorithmic trading and high-frequency environments increasingly dominate global finance, relying solely on constant volatility exposes portfolios to mispriced tail risks. This study addresses this critical gap by exploring an extension of the Black-Scholes framework that integrates stochastic volatility and jump-diffusion models to better reflect the true nature of financial market dynamics.

2. Research Question

How does the incorporation of jump-diffusion processes and stochastic volatility into the Black-Scholes pricing model improve the predictive accuracy of European options pricing, and how can advanced analytical techniques optimize the computational evaluation of these models?

3. Literature Context

Since the foundational work by Black and Scholes (1973), numerous extensions have been proposed to resolve the model's limitations. Merton (1976) introduced the jump-diffusion model to account for discontinuous asset price trajectories, while Heston (1993) developed a stochastic volatility model to capture the mean-reverting nature of volatility. Recent literature emphasizes the necessity of combining both approaches to accurately model short-term pricing dynamics. However, achieving closed-form solutions for these complex models remains computationally intensive. Current research often relies on heavy numerical approximations, highlighting a need for optimized analytical pathways to streamline the evaluation of characteristic functions derived from the underlying partial differential equations (PDEs).

4. Proposed Methodology

This research will employ a rigorous analytical and econometric modeling approach. Initially, the standard Black-Scholes PDE will be extended to incorporate a Poisson jump process. Evaluating the expected payoff under the risk-neutral measure for this extended framework requires computing complex definite integrals over infinite domains.

To handle the intricate characteristic functions associated with the stochastic volatility component, the methodology will involve advanced calculus techniques. Specifically, trigonometric substitutions and contour integration will be utilized to simplify and analytically evaluate these complex definite integrals, thereby deriving robust closed-form approximations. Following the analytical derivation, empirical validation will be conducted using high-frequency historical tick data from the S&P 500 options market. Statistical modeling will be employed to compare the pricing errors of the standard Black-Scholes model against the proposed extended framework.

5. Expected Contribution

By integrating advanced mathematical techniques with financial theory, this research aims to enhance the computational efficiency and accuracy of complex option pricing models. The proposed analytical approach to evaluating the requisite integrals is expected to reduce the reliance on computationally expensive numerical simulations. Ultimately, the findings will provide quantitative analysts and algorithmic traders with a more precise and efficient tool for volatility modeling and risk management, contributing meaningfully to the field of financial engineering.

References

References

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